

CONCEPT 44.5

Hormonal circuits link kidney function, water balance, and blood pressure

In mammals, both the volume and osmolarity of urine are adjusted according to an animal's water and salt balance and its rate of urea production. In situations of high salt intake and low water availability, a mammal can excrete urea and salt in small volumes of hyperosmotic urine with minimal water loss. If salt is scarce and fluid intake is high, the kidney can instead eliminate the excess water with little salt loss by producing large volumes of hypoosmotic urine. At such times, the urine can be as dilute as 70 mOsm/L, less than one-fourth the osmolarity of human blood.

How are urine volume and osmolarity regulated so effectively? As we'll explore in this final portion of the chapter, two major control circuits that respond to different stimuli together restore and maintain normal water and salt balance.

Homeostatic Regulation of the Kidney

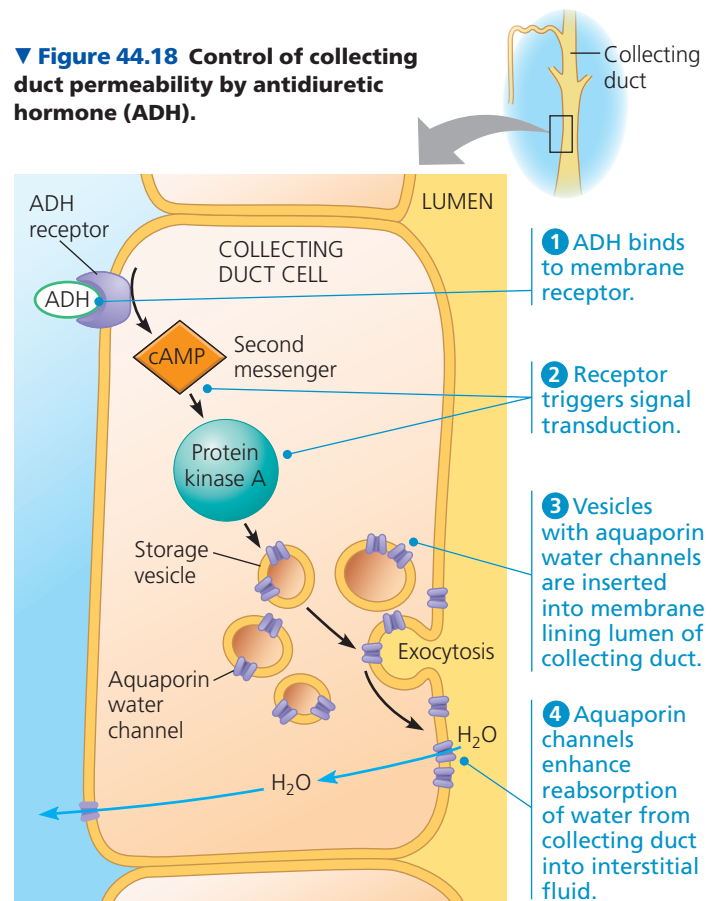
A combination of nervous and hormonal controls manages the osmoregulatory function of the mammalian kidney. Through their effects on the amount and osmolarity of urine, these controls contribute to homeostasis for both blood pressure and blood volume.

Antidiuretic Hormone

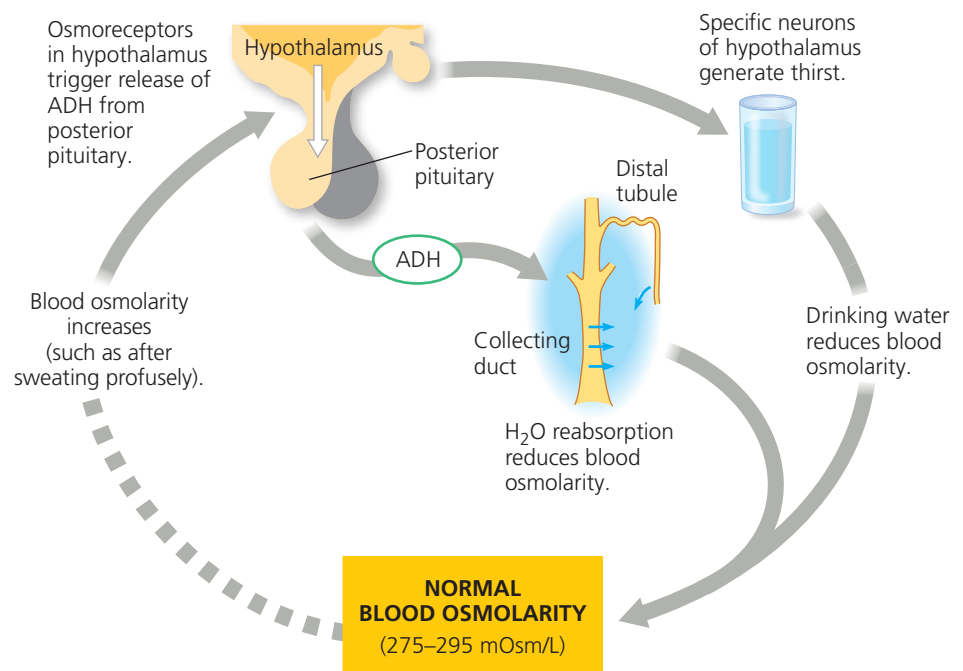
One key hormone in of the kidney is **antidiuretic hormone (ADH)**, also called *vasopressin*. ADH molecules released from the posterior pituitary bind to and activate membrane receptors on the surface of collecting duct cells. The activated receptors initiate a signal transduction cascade that directs insertion of aquaporin proteins into the membrane lining the collecting duct (**Figure 44.18**). More aquaporin channels result in more water recapture, reducing urine volume. (A high level of urine production is called diuresis; ADH is therefore called *anti*diuretic hormone.)

To understand the regulatory circuitry based on ADH, let's consider first what occurs when blood osmolarity rises, such as after eating salty food or losing water through sweating (**Figure 44.19**). When osmolarity rises above the normal range (275–295 mOsm/L), osmoreceptor cells in the hypothalamus trigger increased

▼ **Figure 44.18** Control of collecting duct permeability by antidiuretic hormone (ADH).



▼ **Figure 44.19** Regulation of fluid retention in the kidney. Osmoreceptors in the hypothalamus monitor blood osmolarity via its effect on the net diffusion of water into or out of the receptor cells. When blood osmolarity increases, signals from the osmoreceptors trigger a release of ADH from the posterior pituitary and generate thirst. Water reabsorption in the collecting duct and water intake restore normal blood osmolarity, inhibiting further ADH secretion.



➔ **Mastering Biology Animation: Control of Water Reabsorption**